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SET-1 PROBLEMS;

1.SUM OF TWO MATRIX.

Program:

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r, col, i, j;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &col);
    printf("Enter first matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < col; j++)
            scanf("%d", &a[i][j]);
    printf("Enter second matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < col; j++)
            scanf("%d", &b[i][j]);
    printf("Sum of matrices:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }
```

```
}  
    return 0;  
}
```

Output:

Enter rows and columns: 3 3

Enter first matrix:

2 3 4

5 6 7

8 7 8

Enter second matrix:

7 7 4

8 6 8

5 6 7

Sum of matrices:

9 10 8

13 12 15

13 13 15

2. READ N NUMBER OF VALUES OF VALUES IN AN ARRAY & DISPLAY THEM IN REVERSE ORDER.

Program;

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a[100], n, i;
```

```
    printf("Enter n: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter elements:\n");
```

```
    for(i = 0; i < n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    printf("Reverse order:\n");
```

```
    for(i = n - 1; i >= 0; i--)
```

```
        printf("%d ", a[i]);  
    return 0;  
}
```

Output:

Enter n: 5

Enter elements:

4 4 7 87 56

Reverse order:

56 87 7 4 4

3.COUNT THE TOTAL NUMBER OF DUPLICATE ELEMENTS IN AN ARRAY.

Program:

```
#include <stdio.h>  
  
int main()  
{  
    int a[100], n, i, j, count = 0;  
    printf("Enter n: ");  
    scanf("%d", &n);  
    printf("Enter elements:\n");  
    for(i = 0; i < n; i++)  
        scanf("%d", &a[i]);  
    for(i = 0; i < n; i++)  
    {  
        for(j = i + 1; j < n; j++)  
        {  
            if(a[i] == a[j])  
            {  
                count++;  
                break;  
            }  
        }  
    }  
}
```

```

    }

    printf("Number of duplicate elements = %d", count);

    return 0;
}

```

Output:

Enter n: 5

Enter elements:

768 986 478 768 986

Number of duplicate elements = 2

4.PRINT UNIQUE ELEMENT IN AN ARRAY.

Program:

```

#include <stdio.h>

int main()
{
    int a[100], n, i, j, count;

    printf("Enter n: ");

    scanf("%d", &n);

    printf("Enter elements:\n");

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Unique elements:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(a[i] == a[j])
                count++;
        }

        if(count == 1)
            printf("%d ", a[i]);
    }
}

```

```
    return 0;
}
```

Output:

Enter n: 5

Enter elements:

585 143 489 585 595

Unique elements:

143 489 595

5.INSERT IN SORTED ARRAY.

Program:

```
#include <stdio.h>

int main()
{
    int a[100], n, x, i;
    printf("Enter n: ");
    scanf("%d", &n);
    printf("Enter sorted elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("Enter element to insert: ");
    scanf("%d", &x);
    i = n - 1;
    while(i >= 0 && a[i] > x)
    {
        a[i + 1] = a[i];
        i--;
    }
    a[i + 1] = x;
    n++;
    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
```

```
        printf("%d ", a[i]);  
    return 0;  
}
```

Output:

Enter n: 5

Enter sorted elements:

234 345 456 567 621

Enter element to insert: 792

Array after insertion:

234 345 456 567 621 792

6.INSERT IN UNSORTED ARRAY.

Program:

```
#include <stdio.h>  
  
int main()  
{  
    int a[100], n, x, pos, i;  
    printf("Enter n: ");  
    scanf("%d", &n);  
    printf("Enter elements:\n");  
    for(i = 0; i < n; i++)  
        scanf("%d", &a[i]);  
    printf("Enter position: ");  
    scanf("%d", &pos);  
    printf("Enter element: ");  
    scanf("%d", &x);  
    for(i = n; i >= pos; i--)  
        a[i] = a[i - 1];  
    a[pos - 1] = x;  
    n++;  
    printf("Array after insertion:\n");
```

```

    for(i = 0; i < n; i++)
        printf("%d ", a[i]);
    return 0;
}

```

Output:

Enter n: 5

Enter elements:

567 454 464 567 765

Enter position: 3

Enter element: 345

Array after insertion:

567 454 345 464 567 765

7.MATRIX MULTIPLICATION.

Program:

```

#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;
    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);
    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);
    if(c1 != r2)
    {
        printf("Matrix multiplication not possible");
        return 0;
    }
    printf("Enter first matrix:\n");
    for(i = 0; i < r1; i++)
        for(j = 0; j < c1; j++)

```

```

        scanf("%d", &a[i][j]);
printf("Enter second matrix:\n");
for(i = 0; i < r2; i++)
    for(j = 0; j < c2; j++)
        scanf("%d", &b[i][j]);
for(i = 0; i < r1; i++)
    for(j = 0; j < c2; j++)
    {
        c[i][j] = 0;
        for(k = 0; k < c1; k++)
            c[i][j] += a[i][k] * b[k][j];
    }
printf("Result:\n");
for(i = 0; i < r1; i++)
{
    for(j = 0; j < c2; j++)
        printf("%d ", c[i][j]);
    printf("\n");
}
return 0;
}

```

Output:

Enter rows and columns of first matrix: 2 3

Enter rows and columns of second matrix: 3 2

Enter first matrix:

4 5 5

6 5 5

Enter second matrix:

6 8 7

7 7 9

Result:

94 112

106 128

8.MATRIX TRANSPOSE

Program:

```
#include <stdio.h>

int main()
{
    int a[10][10], r, c, i, j;
    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);
    printf("Enter matrix:\n");
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
    printf("Transpose of matrix:\n");
    for(i = 0; i < c; i++)
    {
        for(j = 0; j < r; j++)
            printf("%d ", a[j][i]);
        printf("\n");
    }
    return 0;
}
```

Output:

Enter rows and columns: 3 3

Enter matrix:

34 54 67

38 29 32

32 39 45

Transpose of matrix:

34 38 32

54 29 39

67 32 45

9.RIGHT DIAGONAL SUM.

Program:

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, sum = 0;
    printf("Enter size of matrix: ");
    scanf("%d", &n);
    printf("Enter matrix:\n");
    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    for(i = 0; i < n; i++)
        sum += a[i][n - i - 1];
    printf("Right diagonal sum = %d", sum);
    return 0; }
```

output:

Enter size of matrix: 3

Enter matrix:

3 4 5

3 7 8

3 6

6

Right diagonal sum = 15

10.CHECK FOR IDENTITY MATRIX.

Program:

```
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, flag = 1;
```

```

printf("Enter size of matrix: ");
scanf("%d", &n);
printf("Enter matrix:\n");
for(i = 0; i < n; i++)
    for(j = 0; j < n; j++)
        scanf("%d", &a[i][j]);
for(i = 0; i < n; i++)
{
    for(j = 0; j < n; j++)
    {
        if(i == j && a[i][j] != 1)
            flag = 0;
        if(i != j && a[i][j] != 0)
            flag = 0;
    }
}
if(flag)
    printf("Identity Matrix");
else
    printf("Not an Identity Matrix");
return 0;
}

```

Output:

Enter size of matrix: 3

Enter matrix:

1 0 0

0 1 0

0 0 1

Identity Matrix

11.TO FIND NUMBER OF VOWELS,CONSONANTS,DIGITS AND SPACES.

Program:

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char str[100];
    int i, vowels = 0, consonants = 0;
    int digits = 0, spaces = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(isalpha(str[i]))
        {
            if(str[i] == 'a' || str[i] == 'e' ||
               str[i] == 'i' || str[i] == 'o' ||
               str[i] == 'u' ||
               str[i] == 'A' || str[i] == 'E' ||
               str[i] == 'I' || str[i] == 'O' ||
               str[i] == 'U')
                vowels++;
            else
                consonants++;
        }
        else if(isdigit(str[i]))
            digits++;
        else if(str[i] == ' ')
            spaces++;
    }
    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("Spaces = %d\n", spaces);
}
```

```
    return 0;
}
```

Output:

Enter a string: my name is bharath i am 18

Vowels = 7

Consonants = 11

Digits = 2

Spaces = 6

12.FREQUENCY OF A CHARACTER IN A STRING.

Program:

```
#include <stdio.h>

int main()
{
    char str[100], ch;
    int i, count = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    printf("Enter character: ");
    scanf("%c", &ch);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
            count++;
    }
    printf("Frequency = %d", count);
    return 0;
}
```

output:

Enter a string: anna university

Enter character: n

Frequency = 3

13.COUNT WORDS IN STRINGS.

Program:

```
#include <stdio.h>

int main()
{
    char str[100];
    int i, words = 0;
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] != ' ' && str[i] != '\n' &&
            (i == 0 || str[i - 1] == ' '))
        {
            words++;
        }
    }
    printf("Number of words = %d", words);
    return 0;
}
```

Output:

Enter a string: i am a student in anna university

Number of words = 7

14.SORT IN ARRAY.

Program:

```
#include <stdio.h>
#include <string.h>
```

```
int main()
{
```

```

char str[10][50], temp[50];
int n, i, j;

printf("Enter number of strings: ");
scanf("%d", &n);

printf("Enter strings:\n");
for(i = 0; i < n; i++)
    scanf("%s", str[i]);

for(i = 0; i < n - 1; i++)
{
    for(j = i + 1; j < n; j++)
    {
        if(strcmp(str[i], str[j]) > 0)
        {
            strcpy(temp, str[i]);
            strcpy(str[i], str[j]);
            strcpy(str[j], temp);
        }
    }
}

printf("Sorted strings:\n");
for(i = 0; i < n; i++)
    printf("%s\n", str[i]);

return 0;
}

```

Output:

Enter number of strings: 5

Enter strings:

bharath
jeno
raja
kishore
Praveen

Sorted strings:

bharath
jeno
kishore
praveen
raja

15.TOGGLE CASE OF A SENTENCE (LOWERCASE TO UPPER CASE VISEVERSA).

Program:

```
#include <stdio.h>
#include <ctype.h>
int main()
{
    char str[100];
    int i;
    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);
    for(i = 0; str[i] != '\0'; i++)
    {
        if(islower(str[i]))
            str[i] = toupper(str[i]);
        else if(isupper(str[i]))
            str[i] = tolower(str[i]);
    }
    printf("Toggled sentence:\n%s", str);
    return 0;
}
```


output:

Enter a sentence: I am a Data Scientist

Toggled sentence:

i AM A dATA sCIENTIST